

Scientific Computation

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Sep. 25, 2025

Lecture 11 - LU Decomposition with Partial Pivoting

Lecture Outline

- **Motivation**
- LU with pivoting

LU and Pivoting

Example 1: The matrix

$$A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

is nonsingular, but does not have LU factorization!

$$l_{21} = \frac{1}{0} = \text{inf}$$

LU and Pivoting cont.

Example 2: The matrix

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

is singular and has the LU factorization:

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}}_L \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}}_U = LU$$

LU and Small Pivots

Example 3: What about the matrix

$$A = \begin{bmatrix} \epsilon & 1 \\ 1 & 1 \end{bmatrix},$$

where $\epsilon > 0$, and especially for $\epsilon \ll 1$?

$$L = \begin{bmatrix} 1 & 0 \\ \frac{1}{\epsilon} & 1 \end{bmatrix}, \quad U = \begin{bmatrix} \epsilon & 1 \\ 0 & 1 - \frac{1}{\epsilon} \end{bmatrix}$$

$$A = LU$$

LU and Small Pivots Cont.

Example 3 cont.: When $\epsilon < \epsilon_{\text{mach}}$, we get $1 - \frac{1}{\epsilon} = -\frac{1}{\epsilon}$ in floating point:

$$\Rightarrow \tilde{U} \approx \begin{bmatrix} \epsilon & 1 \\ 0 & -\frac{1}{\epsilon} \end{bmatrix},$$

And we get $fl(1 - \frac{1}{\epsilon}) = -\frac{1}{\epsilon}$

$$L\tilde{U} = \begin{bmatrix} 1 & 0 \\ \frac{1}{\epsilon} & 1 \end{bmatrix} \begin{bmatrix} \epsilon & 1 \\ 0 & -\frac{1}{\epsilon} \end{bmatrix} = \begin{bmatrix} \epsilon & 1 \\ 1 & \underline{0} \end{bmatrix}$$

$\therefore$ Loss of accuracy

Permute the Rows!

Example 3 cont.: Once again, we apply pivoting and permute the rows:

$$\bar{A} \triangleq \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \epsilon & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ \epsilon & 1 \end{bmatrix},$$

And we get

and we get

$$\bar{L} = \begin{bmatrix} 1 & 0 \\ \varepsilon & 1 \end{bmatrix}, \bar{U} = \begin{bmatrix} 1 & 1 \\ 0 & 1-\varepsilon \end{bmatrix}$$

$\xrightarrow{\bar{L}} f(1-\varepsilon) = 1$

and there's way less error.

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Partial Pivoting Recap

If a pivot is small, the roundoff errors will be multiplied by $1/\text{pivot}$, which is **large**, so at each step of Gaussian elimination choose the row that maximizes the magnitude of the pivot

```
using LinearAlgebra
A = [eps() 1
      1    1]
factorization = lu(A)
println("A: $(A)")
println("L: $(factorization.L)")
println("U: $(factorization.U)")
println("Permutation matrix: $(factorization.P)")
✓ 0.0s

A: [2.220446049250313e-16 1.0; 1.0 1.0]
L: [1.0 0.0; 2.220446049250313e-16 1.0]
U: [1.0 1.0; 0.0 0.9999999999999998]
Permutation matrix: [0.0 1.0; 1.0 0.0]
```

Julia's built-in `lu()` function uses pivoting (see the returned permutation matrix)

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Lecture Outline

- Motivation
- LU with pivoting

Row Swapping

Recall row-swapping from earlier:

e.g. $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$$\begin{bmatrix} \text{---} a_1^T \text{---} \\ \text{---} a_2^T \text{---} \\ \text{---} \vdots \text{---} \\ \text{---} a_n^T \text{---} \end{bmatrix} \mapsto \begin{bmatrix} \text{---} a_2^T \text{---} \\ \text{---} a_1^T \text{---} \\ \text{---} \vdots \text{---} \\ \text{---} a_n^T \text{---} \end{bmatrix}, \quad \Leftrightarrow \pi_k A$$

which is equivalent to left multiplication by a special type of permutation matrix:

$$(\pi_k)_{ij} \triangleq \begin{cases} 1, & i=k, j=l \\ 1, & i=l, j=k \\ 1, & i=j, \text{ and } i \neq l \end{cases}$$

permutation matrix

$\pi_k^{-1} = \pi_k$ (one step)

1 0 ELSB

$\pi_k = 1 \rightarrow$ row swap

$$\pi_k^{-1} = \pi_k$$

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LU with Partial Pivoting

Previously, we derived the LU decomposition by describing Gaussian elimination as the following sequence of elementary matrix multiplications:

$$A^{(k)} = \underbrace{M_k M_{k-1} \cdots M_2 M_1}_{L^{-1}} A. \quad k=1 \cdots n-1 \quad A \in \mathbb{R}^{n \times n}$$

If we include partial pivoting, we will have row-swaps Π_k interleaved with row subtractions M_k :

$$A^{(k)} = M_k \Pi_k M_{k-1} \Pi_{k-1} \cdots M_1 \Pi_1$$

$\hookrightarrow$ pivot first, then subtract the pivot row

Not intuitive to $LU = A$

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Where is the LU structure?

We are actually still computing an LU decomposition, just for the matrix $PA = LU$, where

$$P = \Pi_{n-1} \Pi_{n-2} \cdots \Pi_2 \Pi_1.$$

$$\text{Let } \tilde{M}_k = (\pi_{n-1} \pi_{n-2} \cdots \pi_{k+1}) M_k (\pi_{k+1} \pi_{k+2} \cdots \pi_{n-1}) \quad \text{change base}$$

$$= P_k M_k P_k^{-1}, \quad P_k \triangleq \tau_{n-1} \tau_{n-2} \cdots \tau_{k+1}, \quad P_{n-1} = I$$

$$\tilde{M}_{n-1} \tilde{M}_{n-2} \cdots \tilde{M}_1 = (P_{n-1} M_{n-1} P_{n-1}^{-1}) (P_{n-2} M_{n-2} P_{n-2}^{-1}) \cdots (P_1 M_1 P_1^{-1})$$

Note that: $P_k^{-1} P_{k-1} = (\tau_{k+1} \tau_{k+2} \cdots \underbrace{\tau_{n-1}}_I) (\tau_{n-1} \cdots \tau_k)$

$$= \tau_k$$

$$\therefore \tilde{M}_{n-1} \cdots \tilde{M}_1 = M_{n-1} \tau_{n-1} \cdots M_1 P_1^{-1}$$

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LU Structure Cont.

So we have

$$\tilde{M}_{n-1} \tilde{M}_{n-2} \cdots \tilde{M}_1 = M_{n-1} \tau_{n-1} \cdots M_1 P_1^{-1}$$

Note that $P_1 = \tau_{n-1} \tau_{n-2} \cdots \tau_2$, $P_1^{-1} = \tau_2 \tau_3 \cdots \tau_{n-1}$

$$\begin{aligned} \Rightarrow \tilde{M}_{n-1} \cdots \tilde{M}_1 (PA) &= M_{n-1} \tau_{n-1} \cdots M_1 P_1^{-1} (PA) \\ &= M_{n-1} \tau_{n-1} \cdots M_1 \tau_1 A \end{aligned}$$

But what does $\tilde{M}_k$ look like for each $\tilde{M}_k$?

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LU Structure Cont.

But what does each $\tilde{M}_k$ look like?

$$M_k = \begin{bmatrix} 1 & & & & & \\ & \ddots & & & & \\ & & 1 & & & \\ & & -l_{k+1,k} & & & \\ & & -l_{k+2,k} & & & \\ & 0 & & \ddots & & \\ & & & & 1 & \\ & & & & & -l_{n,k} & \\ & & & & & & 1 \end{bmatrix} = I - l_k e_k^T$$

where $l_k = \begin{bmatrix} 0_k \\ l_{k+1,k} \\ \vdots \\ l_{n,k} \end{bmatrix}$, and $e_k = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix} \rightarrow k\text{th element}$

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LU Structure Cont.

So we have that

$$\begin{aligned} \tilde{M}_k &= P_k \overbrace{(I - l_k e_k^T)}^{M_k} P_k^{-1} \\ &= P_k I P_k^{-1} - (P_k l_k)(e_k^T P_k^{-1}) \\ &= I - P_k l_k (e_k^T P_k^{-1}) \end{aligned}$$

But $P_k = \pi_{n-1} \cdots \pi_{k+1}$, and each π_i only swaps row i
with a row $j \in \{i+1, n\}$

$$\begin{bmatrix} I_{k \times k} & 0 \\ & \tau \end{bmatrix}$$

$$\text{so } e_k^T P_k^{-1} = [0 \ 0 \ \dots \ 1 \ 0 \ 0 \ \dots \ 0] \begin{bmatrix} & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ & & & & & \\ 0 & & & & & \bar{P}_k^{-1} \end{bmatrix} = e_{k'}$$

$$\therefore \tilde{M}_k = I - \underbrace{P_k l_k e_k^T}_{\rightarrow \text{so just swap the elements of } l_k \text{ as you proceed.}}$$

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A Small Example

Example 5: Consider the case where $n = 4$, we compute:

$$M_3 \Pi_3 M_2 \Pi_2 M_1 \Pi_1 A$$

$$\Rightarrow \tilde{M}_3 = M_3, \quad \tilde{M}_2 = \underbrace{\pi_3 M_2 \pi_3}_{\substack{\downarrow \\ P_2}}, \quad \tilde{M}_1 = \underbrace{\pi_3 \pi_2 M_1 \pi_2 \pi_3}_{\substack{\downarrow \\ P_1}}$$

$$\begin{aligned} \therefore \tilde{M}_3 \tilde{M}_2 \tilde{M}_1 P &= M_3 (\pi_3 M_2 \pi_3) (\pi_3 \pi_2 M_1 \pi_2 \pi_3) \pi_3 \pi_2 \pi_1 \\ &= M_3 \pi_3 M_2 \pi_2 M_1 \pi_1 \end{aligned}$$

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Efficiently Storing the Permutation

Finally, we can encode the permutation $P = \Pi_{n-1} \dots \Pi_1$ as follows:

$$P = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 1 & 4 & 3 \end{pmatrix}$$

At each step k , record the index swapped with $\max$ r
as $\text{swap}(k) \in \{k+1, \dots, n\}$
↳ array with $n-1$ elements

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Assignment 1 Hints

1. Carefully read the questions and function signatures!
2. Make your own tests
3. You can store your L and U matrices in a copy of the input matrix A that gets updated at each step $k = 1, 2, \dots, n-1$

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Note: 做 $\widetilde{M}_k = P_k M_k P_k^{-1}$ 的原因是因为我们进行了row change, 所以我们不能直接把变换写成类似 $LU=A$ 的形式, 因为交换行之后, L 不一定是一个lower triangular matrix, 但是我们还是想让 L 保持这个结构, 所以我们使用change base, 让 $\widetilde{M}_{n-1} * \cdots * \widetilde{M}_1$ 来表示 L^{-1} (这是因为我们对 A 进行了row change, $P = \pi_{n-1} * \cdots * \pi_1$, 因此我们可以写成 $PA=LU$, 而 L 是用于在变换以后的vector space表示的。

人话:

1. 先观察每一列, 找出最大的列项作为该列的pivot, 然后找到对应的 Pi matrix进行换行操作。
2. 换行之后, 正常进行elimination找出 L , 注意, 如果此时 L 不是 I (identity matrix), 那么也要进行相应的换行, 具体步骤为: 对角不变, 只换行下半部分的 L 。

例题

设

$$A = \begin{bmatrix} 0 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & 1 & 1 \end{bmatrix}, \quad \text{要做 } PA = LU.$$

步骤 1: 第 1 列选主元并消元

第 1 列绝对值最大在第 3 行 (2), 交换 $r_1 \leftrightarrow r_3$:

$$A_1 = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 2 & 1 \end{bmatrix}.$$

消元系数:

$$m_{21} = \frac{1}{2}, \quad m_{31} = 0.$$

行消元:

$$r_2 \leftarrow r_2 - \frac{1}{2}r_1 \Rightarrow [0, \frac{1}{2}, -\frac{1}{2}], \quad r_3 \leftarrow r_3 - 0 \cdot r_1 \Rightarrow [0, 2, 1].$$

步骤 2: 第 2 列选主元并消元

在行 2 ~ 3 的第 2 列里, $|2| > |\frac{1}{2}|$, 交换 $r_2 \leftrightarrow r_3$:

$$A_2 = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & \frac{1}{2} & -\frac{1}{2} \end{bmatrix}.$$

交换发生在第 2 列之前，因此需要把**已产生的**第 1 列乘子一起交换（下面的 L 会体现）。

消元系数：

$$m_{32} = \frac{\frac{1}{2}}{2} = \frac{1}{4}.$$

行消元：

$$r_3 \leftarrow r_3 - \frac{1}{4}r_2 \Rightarrow [0, 0, -\frac{3}{4}].$$

于是得到

$$U = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & -\frac{3}{4} \end{bmatrix}.$$

组装 P, L

- 两次交换分别是 S_{13} （换 1、3 行）和 S_{23} （换 2、3 行），
故

$$P = S_{23}S_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

- 乘子（考虑第二次交换对第 1 列乘子的行位次影响）：

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ \frac{1}{2} & \frac{1}{4} & 1 \end{bmatrix}.$$

验证：

$$PA = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix}, \quad LU = \begin{bmatrix} 2 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 1 & 0 \end{bmatrix} (= PA).$$

成立： $PA = LU$ 。

用 $PA = LU$ 解方程 $Ax = b$

取 $b = \begin{bmatrix} 3 \\ 1 \\ 2 \end{bmatrix}$.

1. 先算 Pb :

$$Pb = \begin{bmatrix} 2 \\ 3 \\ 1 \end{bmatrix}.$$

2. 解前代 $Ly = Pb$:

$$\begin{cases} y_1 = 2, \\ y_2 = 3, \\ y_3 = 1 - \frac{1}{2}y_1 - \frac{1}{4}y_2 = 1 - 1 - 0.75 = -0.75. \end{cases}$$

3. 解回代 $Ux = y$:

$$\begin{cases} -\frac{3}{4}x_3 = -0.75 \Rightarrow x_3 = 1, \\ 2x_2 + x_3 = 3 \Rightarrow x_2 = 1, \\ 2x_1 + x_2 + x_3 = 2 \Rightarrow x_1 = 0. \end{cases}$$

解得

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解得

$$x = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

快速检验: $Ax = [3, 1, 2]^T = b$ 。